Enterprise RESTful API Documentation

AI-Powered Restaurant Operations Assistant — Complete Endpoint Reference Manual

1. API Specification Overview

Base URL: http://127.0.0.1:8000 | Version: 1.0.0 | Protocols: HTTP/1.1 REST (JSON) | Interactive Docs: /docs.
All endpoints enforce strict Pydantic v2 schema contracts, asynchronous ASGI processing via FastAPI, and domain security guardrails.

2. Master Endpoint Directory

Endpoint	Method	Category	Description
/api/v1/analyze-query	POST	Analyst	Full executive operational query diagnostics with RAG context and root causes.
/api/v1/query	POST	RAG	Grounded operational Q&A powered by Google Gemini 2.5 with citations.
/api/v1/agent/workflow	POST	Agents	Executes 5-agent LangGraph pipeline (Classifier, Root Cause, SOP, Impact, Judge).
/api/v1/analyze	POST	Root Cause	Maps operational symptoms to failure mechanisms and review evidence.
/api/v1/prioritize	POST	Priority	Returns multi-factor ranked complaints filtered by tier, store, and category.
/api/v1/search	POST	Retrieval	Hybrid search fusing dense ChromaDB vectors and sparse BM25 indices (RRF k=60).
/api/v1/forecast	POST	Forecast	Facebook Prophet time-series trend projections (30, 90, 180-day horizons).
/api/v1/similar-restaurants	POST	Benchmark	Discovers operational twin restaurants sharing similar friction profiles.
/api/v1/customer-satisfaction	POST	Analytics	Calculates Customer Satisfaction Index (CSI) and churn propensity tiers.
/api/v1/chat/history	GET/POST/ DEL	History	Cross-session persistent inquiry storage and retrieval in SQLite.
/api/v1/admin/users	CRUD	Admin	Full user management lifecycle (List, Create, Update, Delete).

3. Core Endpoint Specifications (Request & Response Samples)

POST /api/v1/analyze-query
• *Request Format:* {'query': str, 'k': int}
• *Sample Request:* {'query': 'Why are Friday delivery orders cold?', 'k': 10}
• *Sample Response:* {'status': 'success', 'main_issue': {'topic': 'Cold Delivery'}, 'severity': {'score': 4.2}, 'priority': {'tier': 'High', 'sla_hours': 2}, 'recommended_actions': [{'title': 'Thermal Sleeves'}]}

POST /api/v1/agent/workflow
• *Request Format:* {'complaint_text': str, 'restaurant_name': str, 'business_id': str}
• *Sample Request:* {'complaint_text': 'Chicken was raw and manager argued with us', 'restaurant_name': 'Bistro'}
• *Sample Response:* {'category': 'Food Safety', 'severity_score': 4.9, 'priority_tier': 'Critical', 'judge_verdict': 'APPROVED', 'revenue_risk_pct': 35.0, 'operational_urgency': 'P1 - Immediate GM Action'}

POST /api/v1/search
• *Request Format:* {'query': str, 'k': int, 'category': str}
• *Sample Request:* {'query': 'truffle fries soggy', 'k': 3, 'category': 'Food Quality'}
• *Sample Response:* {'query': 'truffle fries soggy', 'total_results': 3, 'results': [{'id': 'rev_7731', 'score': 0.0318, 'text': '...'}]}

POST /api/v1/forecast
• *Request Format:* {'category': str, 'metric': str, 'horizon_days': int, 'freq': str}
• *Sample Request:* {'category': 'Waiting Time', 'metric': 'complaint_count', 'horizon_days': 90}
• *Sample Response:* {'model': 'Prophet', 'trend_direction': 'Surging', 'pct_change': 31.0, 'projected_peak': {'date': '2026-11-27', 'value': 24.1}}